Rafael Fernandes Lopes

Senior Full-Stack Software Engineer

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SUMMARY

Senior full-stack software engineer with 25+ years building scalable web, mobile and cloud products with TypeScript, React, Next.js, Node.js, Python and Java (Spring) on AWS and GCP. At Britannica Education, built the SSO platform behind 15 products and 5.6M+ student accounts and helped launch an award-winning AI teaching workspace; previously co-founded an EdTech startup that reached R$490k ARR and spent 16 years as a Computer Science professor (Ph.D.) and innovation director.

EXPERIENCE

Senior Software Engineer · Fullstack Labs

Mar 2022 – Present

Remote · Clients: Britannica Education (May 2022 – Present); Fueled (Mar 2022 – May 2022)

- Unified single sign-on (SSO) for 15 products, serving 266K+ institution and 5.6M+ student accounts (1.4M+ users logged in since 2024), by building the Next.js/TypeScript Education Portal from scratch and migrating 4 legacy Java products onto it in 2 months.
- Helped launch Britannica Studio, an AI teaching workspace (2026 Tech & Learning Best of Show and EdTech Breakthrough AI Teaching Assistant of the Year; 780+ institutions in 100+ countries), by setting up its Next.js frontend and authentication; also built Ask Britannica's streaming AI chat UI in React.
- Cut Britannica Escola's homepage API calls by 67% and database queries by 33% with batch endpoints in its Python AWS Lambda backend, and connected that backend to Britannica's Oracle-based authentication services over Avro RPC.
- Stopped users from being signed into the wrong institutional accounts by tracing a GeoIP flaw to the CDN's IP and moving the check to a JWT-protected endpoint; one of 227 critical and high-priority bugs resolved.
- Migrated the portal from Netlify to AWS ECS Fargate for autoscaling and higher availability by writing its Terraform and a CircleCI Docker deployment pipeline.
- Raised the portal's line coverage to 95% (1,099 Jest and React Testing Library tests) and built an AI coding-agent harness for code review and test writing; also fixed WCAG 2 AA accessibility issues in the portal, Escola, ImageQuest and Britannica School.

Co-founder & Lead Developer · ImaginaKIDS

Oct 2017 – Mar 2022

EdTech startup where students write and illustrate their own books, printed as real books

- Grew the company from zero to R$490k ARR with under 10% churn, reaching 15k+ students, 300+ teachers and 380k+ pages created, by leading product and engineering as co-founder and managing 2 developers and a UX/UI designer.
- Built 6 products from scratch, owning the architecture: a React Native iPad app (Swift gestures), a React web editor, a Flutter microlearning app, a multi-payment e-commerce store, a Node.js book-to-PDF service (Bull) and a Django print-partner app.

Associate Professor, Researcher & Innovation Director · UFMA and IFMA

2006 – 2022

Federal Institute of Maranhão (IFMA) 2006–2016; Federal University of Maranhão (UFMA) 2016–2022, alongside ImaginaKIDS

- Helped 7 of 10 startups win public funding by mentoring their founders as UFMA's director of entrepreneurship and innovation, and managed and advised dozens of R&D software projects (10+ funded) in web, mobile and distributed systems.
- Taught Computer Science to hundreds of students; supervised 7+ Master's students, published 6 journal articles and 48+ conference papers, and peer-reviewed for IEEE and Springer.

Web Developer · UFMA IT Center and Fapema

2003 – 2006

- Built and maintained the university's and Fapema's websites and information systems in PHP, MySQL and Java EE (JSP, JSF).

SKILLS

- **Languages:** TypeScript, JavaScript, Python, Java, Dart, SQL, HTML, CSS
- **Frontend and mobile:** React, Next.js (SSR/SSG), Redux Toolkit, Tailwind CSS, Sass, React Native, Flutter, WCAG 2 AA
- **Backend and APIs:** Node.js, Express, Java (Spring), Django, Serverless (AWS Lambda), REST APIs, SSO, OAuth, JWT, gRPC, WebSocket, RabbitMQ, MQTT
- **Databases:** PostgreSQL, MongoDB, Redis, DynamoDB, Firebase
- **Cloud and DevOps:** AWS (ECS Fargate, Lambda, CloudWatch), GCP, Terraform, Docker, CI/CD (GitHub Actions, CircleCI)
- **Testing and practices:** Jest, Vitest, React Testing Library, code review, application security, performance optimization, AI coding agents

PROJECTS

- **Simula Finanças** (simulafinancas.com.br): Next.js finance tools built with AI assistance; 100 in Lighthouse SEO and accessibility.

EDUCATION

- **Ph.D. in Electrical Engineering**, Federal University of Campina Grande (UFCG), 2013
- **M.Sc. in Electrical Engineering**, Federal University of Maranhão (UFMA), 2006
- **B.Sc. in Computer Science**, Federal University of Maranhão (UFMA), 2003